A QUASI-COMPACT QUASI-SEPARATED ALGEBRAIC SPACE ADMITS A SURJECTIVE INTEGRAL COVER BY A SCHEME FACTORING LOCALLY THROUGH A GIVEN ETALE COVER

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Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Bounded fibres of the etale cover. lemma-bounded-fibres (carded L26703). The degree bound d for U -> X.
- Degree stratification. lemma-quasi-compact-reasonable-stratification (carded L26710). The degree of the fibres of a quasi-compact etale morphism does not go up under specialization.
- Integral extension of the cover domain. lemma-extend-integral-morphism (same chapter, carded L26059). Factorization Z -> Z' -> Y with Z' -> Y integral.
- Normalization in a quasi-finite separated morphism. Morphisms of Spaces, lemma quasi-finite-separated-quasi-affine (cross-chapter, named). The factorization U -> Y -> X with pi integral.
- Scheme-theoretic density and integral composition. Morphisms of Spaces, lemmas semi-diagonal and composition-integral (cross-chapter, named). The decomposition U x_X Y = U amal W and integrality of the assembled morphism.

Source reference. Original lemma 03JX: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 952–952. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G2D: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2002–2002. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07S8: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1653–1653. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ABS: focused-seeds/geometry/S9-decent-there-is-a-scheme-integral-over/ref-tags.txt, source lines 5–5. Navigation pointer only:

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03KO: focused-seeds/geometry/S9-decent-there-is-a-scheme-integral-over/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S.*

(1) *There exists a surjective integral morphism $Y \to X$ where Y is a scheme,*
(2) *given a surjective étale morphism $U \to X$ we may choose $Y \to X$ such that for every $y \in Y$ there is an open neighbourhood $V \subset Y$ such that $V \to X$ factors through U.*

Proof. Part (1) is the special case of part (2) where $U = X$. Choose a surjective étale morphism $U' \to U$ where U' is a scheme. It is clear that we may replace U by U' and hence we may assume U is a scheme. Since X is quasi-compact, there exist finitely many affine opens $U_i \subset U$ such that $U' = \coprod U_i \to X$ is surjective. After replacing U by U' again, we see that we may assume U is affine. Since X is quasi-separated, hence reasonable, there exists an integer d bounding the degree of the geometric fibres of $U \to X$ (see Lemma 03JX). We will prove the lemma by induction on d for all quasi-compact and separated schemes U mapping surjective and étale onto X. If $d = 1$, then $U = X$ and the result holds with $Y = U$. Assume $d > 1$.

We apply Morphisms of Spaces, Lemma 0ABS and we obtain a factorization

$$\begin{array}{ccc} U & \xrightarrow{\quad j \quad} & Y \\ & \searrow \quad \swarrow_{\pi} & \\ & X & \end{array}$$

with π integral and j a quasi-compact open immersion. We may and do assume that $j(U)$ is scheme theoretically dense in Y. Then $U \times_X Y$ is a quasi-compact, separated scheme (being integral over U) and we have

$$U \times_X Y = U \amalg W$$

Here the first summand is the image of $U \to U \times_X Y$ (which is closed by Morphisms of Spaces, Lemma 03KO and open because it is étale as a morphism between algebraic spaces étale over Y) and the second summand is the (open and closed) complement. The image $V \subset Y$ of W is an open subspace containing $Y \setminus U$.

The étale morphism $W \to Y$ has geometric fibres of cardinality $< d$. Namely, this is clear for geometric points of $U \subset Y$ by inspection. Since $|U| \subset |Y|$ is dense, it holds for all geometric points of Y by Lemma 07S8 (the degree of the fibres of a quasi-compact étale morphism does not go up under specialization). Thus we may apply the induction hypothesis to $W \to V$ and find a surjective integral morphism $Z \to V$ with Z a scheme, which Zariski locally factors through W. Choose a factorization $Z \to Z' \to Y$ with $Z' \to Y$ integral and $Z \to Z'$ open immersion

(Lemma 0G2D). After replacing Z' by the scheme theoretic closure of Z in Z' we may assume that Z is scheme theoretically dense in Z'. After doing this we have $Z' \times_Y V = Z$. Finally, let $T \subset Y$ be the induced closed subspace structure on $Y \setminus V$. Consider the morphism

$$Z' \amalg T \longrightarrow X$$

This is a surjective integral morphism by construction. Since $T \subset U$ it is clear that the morphism $T \to X$ factors through U. On the other hand, let $z \in Z'$ be a point. If $z \notin Z$, then z maps to a point of $Y \setminus V \subset U$ and we find a neighbourhood of z on which the morphism factors through U. If $z \in Z$, then we have an open neighbourhood of z in Z (which is also an open neighbourhood of z in Z') which factors through $W \subset U \times_X Y$ and hence through U. $\square$